

Homework #2

Space Missions and Systems

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1 Part 1

We can divide attitude determination methods in two main categories:

- **Static determination approaches**

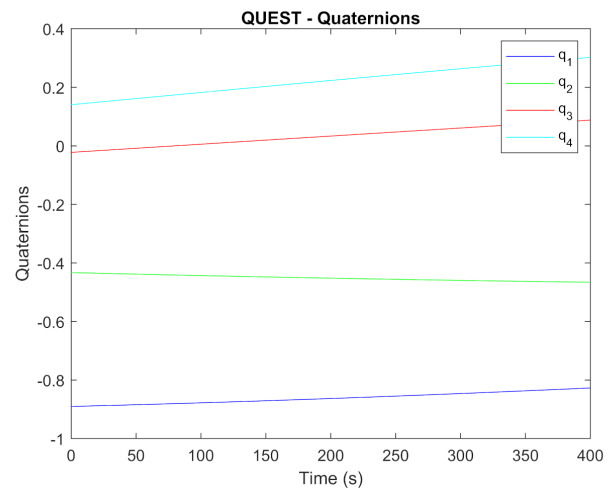
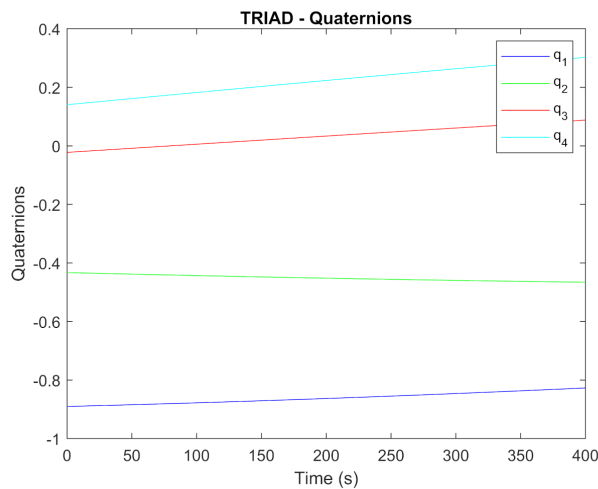
These act at the same time the measurements are taken, or very close to that instant in such a way that the spacecraft motion between the measurements can be ignored.

- **Filtering processes**

These methods have a memory of past measurements, and can thus generally provide a more accurate attitude estimate than static methods. The drawback is requiring a priori estimate of the state.

Three methods are here used to estimate the attitude of the spacecraft in a time interval $[0s, 400s]$.

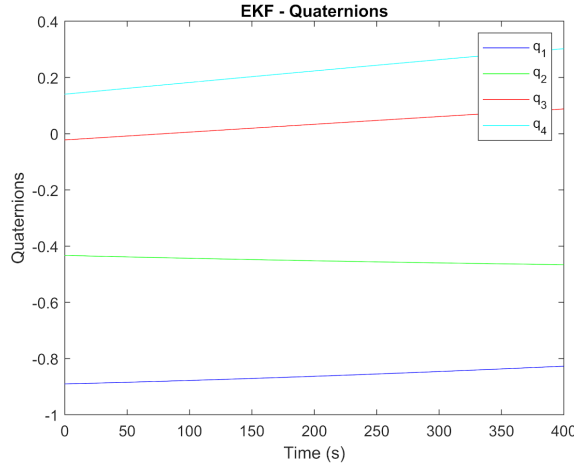
The first method is the TRIAD algorithm, a deterministic approach that incorporates just enough observation information to determine the attitude. The TRIAD algorithm uses only three out of the four pieces of information obtainable from two measured vectors to compute an attitude solution, from here onwards expressed as a quaternion. Since only two vectors at a time can be used in the TRIAD algorithm, it is recommended to use the two vectors that are mostly orthogonal to ensure a smaller loss of data from the measurements. In the following picture the second and fourth measurements are taken into account for the first and fourth observation times $[0s, 100s]$, $[300s, 400s]$, shifting to the first and third for the third interval $[200s, 300s]$. Lastly in the second interval $[100s, 200s]$ there are not enough measurements to run the algorithm, so a propagation had to be considered.



$$\begin{aligned}\dot{\vec{q}} &= \frac{1}{2}\Omega\vec{q} \\ \dot{\Phi} &= -[\vec{\omega}_{meas} \times] \Phi \\ P_{k+1} &= \Phi P_k \Phi^T\end{aligned}$$

The second method used was the QUEST algorithm, a generalization of the TRIAD, as it allows more than two measurements at a time and also considers weighting coefficients. It is preferred to other estimators for its efficiency, as it searches for the optimal solution without having to solve the complete eigenvalue problem explicitly.

The QUEST was used as a batch filter as well, without encountering any problems, albeit some adjustments had to be done for the two intervals $[300s, 307.3s)$ and $[365.9s, 367.4s)$, when ST3 lost track of a star, probably due to wandering off the field view of the sensor. Once again, in the second interval the propagation was considered to estimate further the attitude.



Lastly, the EKF method defines a state vector, comprising the attitude variables as well as other variables coming from additional systems (i.e. the Farrenkopf model for the gyroscope) and thus can keep track of the bias of the gyroscope. This method requires a priori estimate, given by the previous QUEST, which can then be analyzed and improved upon consideration of the angular velocities of the gyro, while also correcting them. The same problem occurred here in the second interval, where the propagation took the place of the EKF. This time, an improved version of the latter was considered: the matrix Q keeps track of the noise introduced by the gyroscope, and since it cannot be corrected anymore for lack of measurements, Δt increases by $0.1s$ each time a step forward takes place.

$$\begin{aligned}\dot{\vec{q}} &= \frac{1}{2}\Omega\vec{q} \\ \dot{\Theta} &= -[\vec{\omega}_{est} \times] \Theta \\ \dot{\Psi} &= -[\vec{\omega}_{est} \times] \Psi - \frac{1}{2}I_{3 \times 3} \\ \Phi &= \begin{bmatrix} \Theta & \Psi \\ 0_{3 \times 3} & I_{3 \times 3} \end{bmatrix} \\ Q &= \begin{bmatrix} \sigma_v^2 I_{3 \times 3} & 0_{3 \times 3} \\ 0_{3 \times 3} & (\sigma_u \Delta t)^2 I_{3 \times 3} \end{bmatrix} \\ P_{k+1} &= \Phi P_k \Phi^T + Q\end{aligned}$$

Here it's presented a table with the attitude estimation at certain times, as got from the EKF approach.

	$t = 0s$	$t = 100s$	$t = 200s$	$t = 300s$	$t = 400s$
q_1	-0.8901	-0.8777	-0.8630	-0.8460	-0.8269
q_2	-0.4329	-0.4431	-0.4520	-0.4596	-0.4659
q_3	-0.0223	0.0057	0.0335	0.0609	0.0879
q_4	0.1404	0.1822	0.2233	0.2634	0.3023

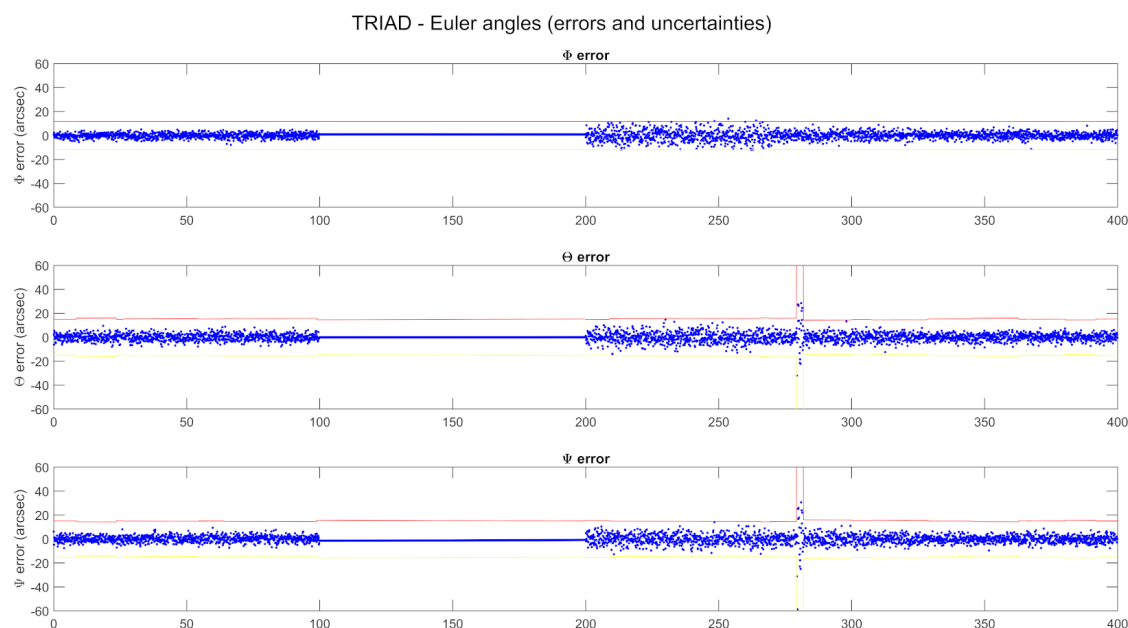
2 Part 2

The three methods are here briskly discussed. As we can see from the following figures, the Euler angles do present an error relative to the true attitude of the spacecraft, represented along their σ bounds. The first interval $[0s, 100s]$ is pretty stable, centered around zero and do not show any interesting behavior, meaning the estimators are working nominally.

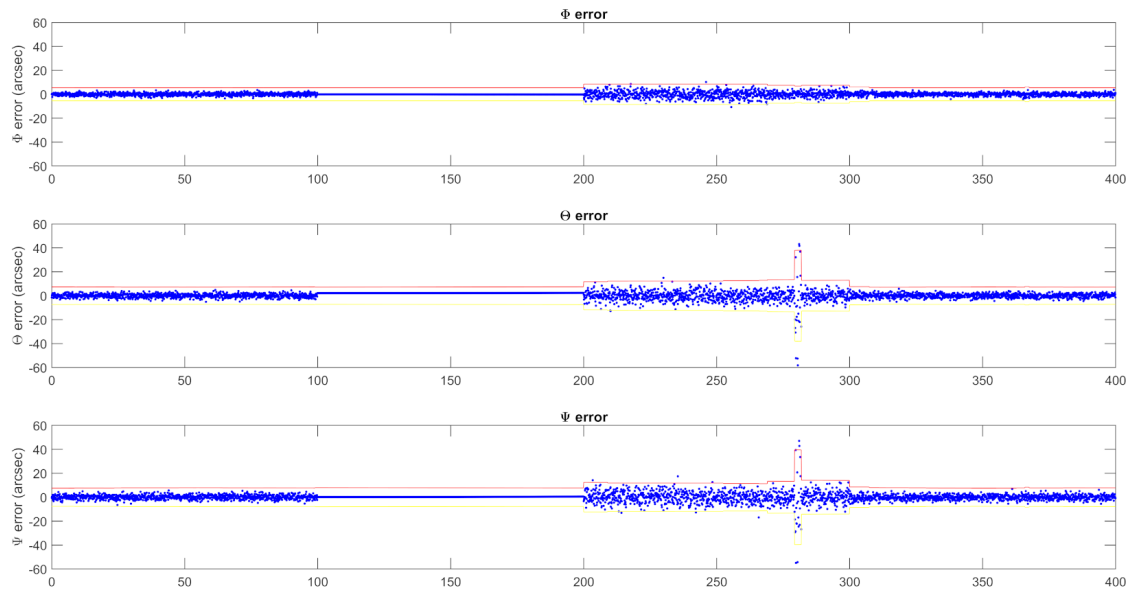
The second interval $[100s, 200s]$ instead presents a major problem, as ST1, as well as ST2 and ST3 stop providing the on-board computer the measurement required to estimate the attitude. As we've discussed before, a propagation by the kinematics equation is considered to solve this inconvenience. The biggest problem with the propagation is certainly the propagation of the σ bounds, as they increase exponentially, while the quaternion and thus the attitude does not appear to change much. This climb can easily be seen in the EKF pertaining figure, while it's not represented in the other figures due to an incomplete model for propagation.

In the third interval $[200s, 300s]$ we come back to a pretty stable configuration. Nonetheless a surge is observed around the mark $280s$, probably due to a loss of data from the measurements. This can happen if said measurements overlap (the vectors tend to be parallel). The EKF, after reducing rapidly the uncertainty interval, solves this problem using the gyroscope as a "backup device".

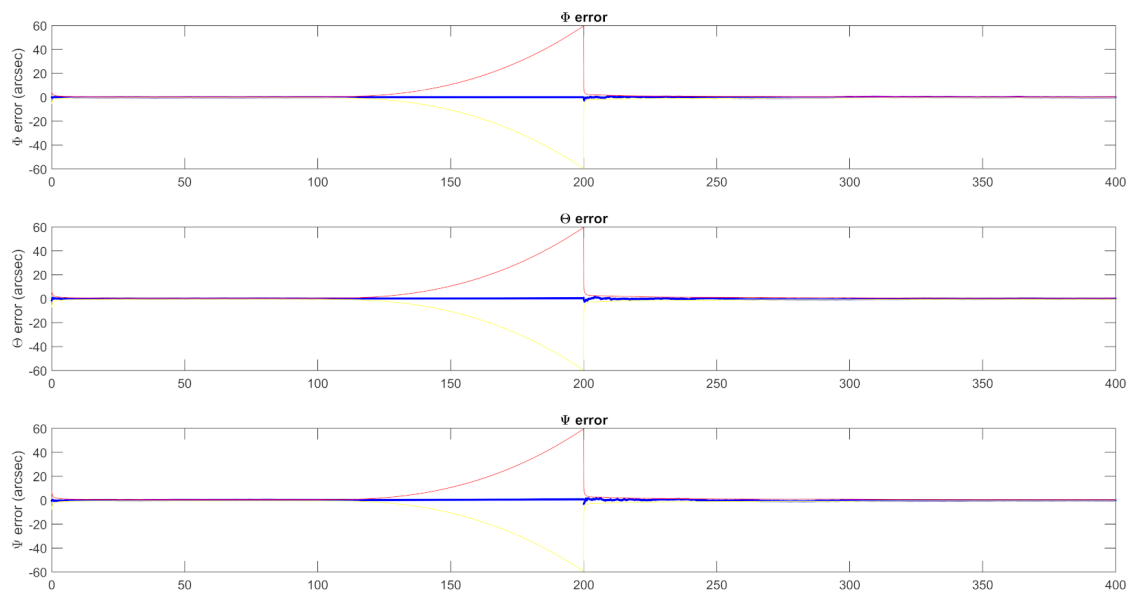
The last interval $[300s, 400s]$ does not present any issues apart the occasional star that disappears from ST3's FOV, thus invalidating that measurement. Once again, the EKF is able to improve the estimated attitude, reducing the uncertainties boundary.



QUEST - Euler angles (errors and uncertainties)

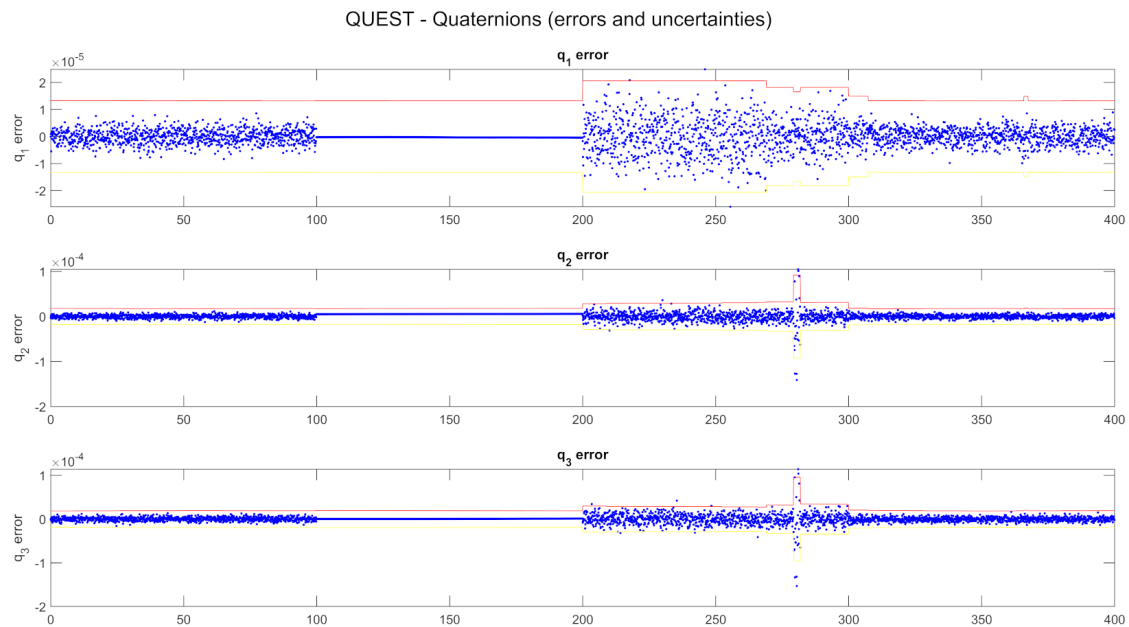
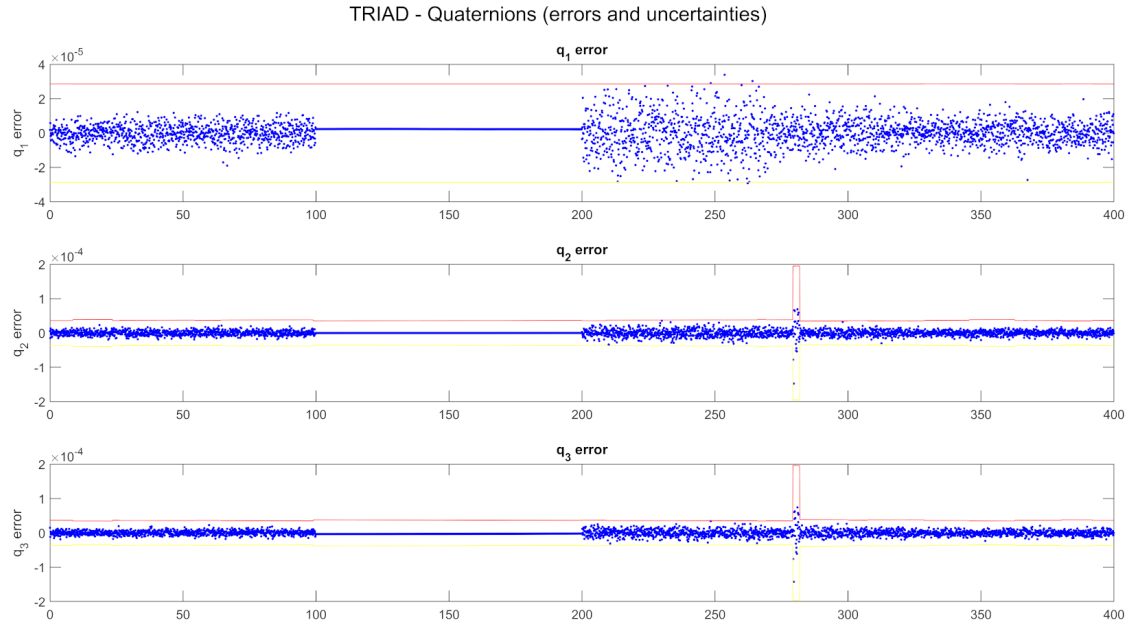


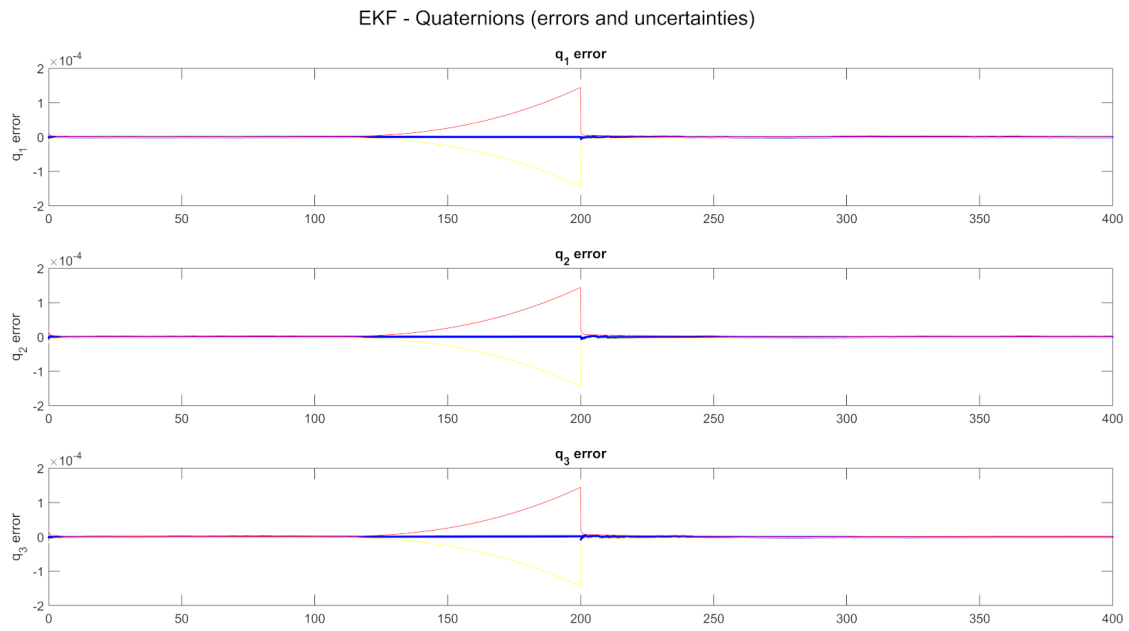
EKF - Euler angles (errors and uncertainties)



3 Part 3

The best algorithm for the attitude determination is definitely the EKF approach, even though it's computationally labor intensive it can estimate the attitude with a smaller degree of uncertainty. It really depends upon the mission requirements, as for moderate accuracy missions the TRIAD, or the QUEST might be considered better suited for their low computational burden. For higher accuracy missions, the EKF is usually the method of choice.





Both the angular velocity error and bias are very small, only having a surge after the propagation interval is over, due to the bias not being estimated anymore and thus the gyroscope wandering freely providing falsified data. The formulas considered are the following:

$$\vec{\beta} = \vec{\omega}_{meas} - \vec{\omega}_{est}$$

$$\Delta\vec{\omega} = \vec{\omega}_{true} - \vec{\omega}_{est}$$

